

A scaled rotation disproves conjecture 00000007174

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The stated antisymmetric-norm lower bound fails even for an invertible normal real matrix. We use the standard spectral (Euclidean operator) condition number $\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2$ and also check the ratio of eigenvalue moduli, covering the source's spectral-ratio wording.

On the actual real Euclidean plane let

$$A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix}.$$

Direct multiplication gives $BA = AB = I$, so $B = A^{-1}$. For every vector $v = (x, y)$,

$$Av = (-2y, 2x), \quad \|Av\|_2 = 2\|v\|_2, \quad \|Bv\|_2 = \frac{1}{2}\|v\|_2.$$

Taking the supremum over unit vectors therefore gives the actual operator norms $\|A\|_2 = 2$ and $\|B\|_2 = 1/2$. Consequently

$$\kappa_2(A) = 1.$$

The adjoint for the standard Euclidean inner product is $A^* = -A$. In particular $A^*A = AA^*$, so A is normal. Its standard antisymmetric part is

$$K = \frac{A - A^*}{2} = A, \quad \|K\|_2 = 2 > 1 = \kappa_2(A).$$

Thus the norm of the antisymmetric part is not a lower bound for the spectral condition number. If “antisymmetric” instead denotes the unnormalized difference $A - A^*$, its norm is 4, so that convention also fails.

For completeness, the complexified matrix has characteristic equation

$$\det(A - zI) = z^2 + 4 = (z - 2i)(z + 2i).$$

Its two eigenvalues are exactly $2i$ and $-2i$, both of modulus 2. The ratio of largest to smallest eigenvalue modulus is therefore also 1, and is again strictly less than the antisymmetric norm 2. The witness disproves the lower-bound inequality itself, independently of the further assertion about attaining the bound.

Formal correspondence. Lean realizes the real Euclidean plane as $\mathbb{C}$ with its actual real Hilbert structure and defines $A(z) = 2iz$, $B(z) = -iz/2$ as actual continuous real linear maps. It verifies the coordinate matrix above, both inverse identities, both actual operator norms, the actual adjoint, normality and the strict failure of the bound. It separately verifies the determinant factorization and complete eigenvalue list of the complexified 2×2 matrix and the modulus ratio. The complex norm is the Euclidean norm; no coordinate supremum norm is used. The project is pinned to Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`.